Parametric Prime Families via Quadratic Forms:
Layer Invariants, Conditional Proofs,

and Connections to the Riemann Zeta Function

Sequel to: *Heuristic Distribution of the Minimal Offset, Large-Scale Numerical Validation, and Statistical Control of Spurious Spectral Correlations* [Paper I, April 2026]

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April 2026 (revised version)

Abstract

We study the parametric family $p = km(m+1) + \varepsilon + 2kq$, with $m \in \mathbb{N}^*$, $\varepsilon \in \{-1, +1\}$, $k \in \{3, 15, 21\}$ fixed, and $|q|$ minimal, which maps each prime to a generalised hexagonal base $H_m^{(k)} = km(m+1) + 1$. This unified paper synthesises three independent research sessions into a single coherent work, covering $N = 10^8$ primes ($k = 3$, $m \in [1, 5773]$) and $N = 10^7$ primes ($k \in \{15, 21\}$).

Main rigorous results (Theorems 2.3 and 3.2). The class ε is uniquely determined by $p \bmod 6$; $|q_{\min}| \leq (m+1)/2$ unconditionally; $\mathbb{E}[q \mid m] = 0$ algebraically; the admissible class count is $\varphi(2k)/2$ (Dirichlet). Under GRH: $|q_{\min}(m, \varepsilon)| = O_k(m \log^2 m)$ (Theorem 2.3, via Heath-Brown's Linnik-type bound). The Pocklington–Lehmer criterion proves primality of $p = 3m(m+1) + 1$ unconditionally for numbers up to $30\,000^+$ decimal digits, exploiting the fully factored structure of $p - 1$. The mean gap satisfies $\bar{\delta}(m) \approx 0.296 \times \ln p(m)$, stable to ± 0.006 over five orders of magnitude (Hardy–Littlewood constant for $f(m) = 3m^2 + 3m + 1$).

Principal heuristic contribution (Conditional Proposition 9.4). The constant $C_0(k) := \langle |q| \rangle / \sqrt{p} = 1/(4\sqrt{k})$ is empirically stable to $< 0.02\%$ across $k \in \{3, 15, 21\}$; universality for all admissible k is conjectured. Under H1 + GRH + H3, the conditional expectation $\mathbb{E}[|q| \mid m] = m/4$ is established in five steps (Conditional Proposition 9.4). The analytical formula $C^*(k) = (\beta + 1)/[4(\beta + 2)\sqrt{k}]$ has error $< 0.5\%$.

Zeta function connections. The Riemann zeros $\rho = \frac{1}{2} + i\gamma$ govern oscillations of n_m via the Riemann–von Mangoldt explicit formula (Level I). Under H1 + GRH + H3: GRH $\Rightarrow$ C1; the converse is not claimed (Level II). The Riemann zeros are *not* spectral frequencies of $Q(r)$: $R^2 = 1.16 \times 10^{-7}$ at $N = 10^8$, three permutation tests (Level IV).

Negative results. The bias $\langle q \rangle \approx +0.4$ is a code artifact. Three conjectures corrected: C1 ($m/2 \to m/4$); C2 (-65%); C3 (divisor). The exponent $\sigma(X(r)) \sim r^{0.275}$ is not confirmed at $N = 10^8$.

MSC 2020: 11N32, 11N13, 11Y11, 11M41, 62P99.

Keywords: prime numbers, quadratic forms, centred hexagonal numbers, Bateman–Horn conjecture, Weyl equidistribution, Riemann zeta function, Pocklington criterion, GRH, Hardy–Littlewood constant.

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1 Introduction

Every prime $p > 3$ satisfies $p \equiv \pm 1 \pmod 6$, placing it at signed distance 1 from the nearest multiple of 6. This elementary observation seeds a richer parametric structure. Writing

$$p = km(m+1) + \varepsilon + 2kq, \qquad m \in \mathbb{N}^*,\ \varepsilon \in \{-1, +1\},\ q \in \mathbb{Z},\ |q| \text{ minimal},$$

one associates each prime to the generalised hexagonal base $H_m^{(k)} := km(m+1)+1$ and measures its signed displacement q from that base. For $k = 3$ the bases $H_m = 3m(m+1)+1$ are the centred hexagonal numbers, elements of $\mathbb{Z}[\omega]$ ($\omega = e^{2\pi i/3}$, integers of $\mathbb{Q}(\sqrt{-3})$), which index atom positions in hexagonal lattices [6].

This paper extends and deepens the results of Paper I [1]:

(i) rigorous Pocklington–Lehmer primality certificates for $p = 3m(m+1)+1$ up to 30 000 decimal digits (Theorem 3.2);

(ii) layer-structure analysis at $N = 10^8$, establishing the intra-layer ratio $\sigma(S_m)/\sqrt{n_m/12} \approx$ 0.63 (Revised Result 6.1);

(iii) empirical universality of $C_0(k) = 1/(4\sqrt{k})$ across $k \in \{3, 15, 21\}$;

(iv) a conditional proof of $\mathbb{E}[|q| \mid m] = m/4$ (Conditional Proposition 9.4) and a four-level map of the connection to $\zeta(s)$.

Epistemic key

Every result carries one of four labels: **Rigorous** (unconditional, from classical theorems); **Heuristic** (conditional on H1 or GRH); **Negative** (explicit invalidation of a prior claim); **Revised** (status corrected by the $N = 10^8$ campaign).

Central contribution of this paper

The two main *rigorous* results are: Theorem 2.3 ($|q_{\min}| = O_k(m \log^2 m)$ under GRH, via the Heath-Brown GRH-conditional Linnik bound) and Theorem 3.2 (unconditional Pocklington–Lehmer primality certificates). The principal *heuristic* contribution is Conditional Proposition 9.4: under H1 + GRH + H3, $\mathbb{E}[|q| \mid m] = m/4$, proved in five explicit steps. All remaining results are supporting evidence or open problems, labelled throughout with their epistemic status.

Relation to Paper I

This paper is a direct sequel to [1], hereafter *Paper I*, which established: (i) the heuristic law $\mathbb{E}[|q_{\min}(k,m)|] \sim \ln m/C_k$ (validated at $N = 10^6$, $k = 21$); (ii) the structural decomposition $Q \approx \alpha\sqrt{p} + \beta D_N$; (iii) the decisive negative result on spectral correlations $Q \leftrightarrow \zeta(s)$.

What this paper adds. Extending the analysis to $N = 10^8$ ($k = 3$) and to the universality across $k \in \{3, 15, 21\}$, we establish the layer invariant $\sigma(S_m)/\sqrt{n_m/12} \approx 0.63$, the constant $C_0(k) = 1/(4\sqrt{k})$, a conditional proof of $\mathbb{E}[|q| \mid m] = m/4$, an analytical formula for $C^*(k)$, rigorous Pocklington–Lehmer primality certificates, and a four-level map of the connection to $\zeta(s)$.

Parameterisation note. Paper I uses the one-sided convention $q \geq 0$ for the branch $\varepsilon = +1$ only (Appendix A of [1]), yielding $\mathbb{E}[q \mid m] = m/2$. *This paper* uses the signed convention $q \in \mathbb{Z}$ for the full family ($\varepsilon = \pm 1$), yielding $\mathbb{E}[|q| \mid m] = m/4$. These are consistent: $m/2 = 2 \times (m/4)$ reflects the doubling when passing from the one-sided to the two-sided parameterisation. There is no contradiction between the two results.

One genuine correction. Theorem D.2 of Paper I states $|q_{\min}| \ll_k m \log m$ under GRH. A careful application of Heath-Brown's GRH-conditional Linnik bound [14] (see Theorem 2.3 below) gives the slightly weaker $O_k(m \log^2 m)$; Paper I's bound should read $O_k(m \log^2 m)$.

2 Theoretical Framework

2.1 Definitions and basic properties

Definition 2.1. Fix $k \geq 1$ with $\gcd(k, 6) = 1$ or $k = 3$. For $m \in \mathbb{N}^*$, $\varepsilon \in \{-1, +1\}$ admissible, $q \in \mathbb{Z}$, define $p = km(m+1) + \varepsilon + 2kq$. The minimal residue is

$$q_{\min}(m, \varepsilon) = \arg\min\Big\{|q| : km(m+1) + \varepsilon + 2kq \in \mathcal{P}\Big\}.$$

The admissible classes are $\varepsilon \in (-k, k] \cap \mathbb{Z}$ with $\gcd(\varepsilon, 2k) = 1$.

Rigorous

Theorem 2.2 (Dirichlet [9]). *For every $m \geq 1$ and every admissible ε, the arithmetic progression $\{km(m+1) + \varepsilon + 2kq : q \in \mathbb{Z}\}$ contains infinitely many primes. Hence $q_{\min}(m, \varepsilon)$ is always well-defined.*

Theorem 2.3 (Bound under GRH). [Rigorous, conditional on GRH] *Assume the Generalised Riemann Hypothesis (GRH) for all Dirichlet L-functions. Then*

$$|q_{\min}(m, \varepsilon)| = O_k(m \log^2 m).$$

Proof. The progression $\mathcal{A}_{k,m,\varepsilon} = \{km(m+1) + \varepsilon + 2k\mathbb{Z}\}$ has modulus $\hat{q} = 2k$ and first term $A = km(m+1) + \varepsilon \sim_k km^2$. By the Heath-Brown GRH-conditional bound on Linnik's problem [14], the least prime $p \equiv A \pmod{\hat{q}}$ with $p > A$ satisfies

$$p \;\le\; A + C(\hat{q})\sqrt{A}\,(\log A)^2$$

for an explicit constant $C(\hat{q})$ (depending only on $\hat{q}$). Since $p = A + 2kq$ and $A \sim_k km^2$,

$$|q| \;\le\; \frac{C(\hat{q})\sqrt{A}\,(\log A)^2}{2k} \;=\; O_k\big(m(\log m)^2\big). \qquad \square$$

Remark 2.4. The heuristic prediction is $|q_{\min}| = O(\log m)$ (Cramér model); the unconditional gap is Open Problem 12.1.

Rigorous

Proposition 2.5 (Admissible classes)**.** *The number of admissible classes is exactly $\varphi(2k)/2$: $k = 3 \Rightarrow 2$ classes; $k = 15 \Rightarrow 8$; $k = 21 \Rightarrow 12$. Each class contains asymptotic density $1/\varphi(2k)$ of the primes (Dirichlet [9]).*

Numerical check. At $N = 10^7$, $k = 15$: each of the 8 classes contains exactly $(1/8) \times 664\,569$ primes, error $< 0.02\%$.

2.2 Geometric reformulation and $O(1)$ algorithm

Setting $T = (p - \varepsilon)/(2k)$ and $t_m = m(m+1)/2$, the family becomes $q = T - t_m$. The optimal index is

$$m_{\text{opt}} = \left\lfloor \frac{-1 + \sqrt{1 + 8T}}{2} \right\rfloor,$$

computable in $O(1)$ per prime after an $O(N \log\log N)$ segmented sieve. For $k = 3$: $H_m = 3m(m+1) + 1 \in \mathbb{Z}[\omega]$; ε is uniquely determined by $p \bmod 6$ for $p > 3$; $|q_{\min}| \le (m+1)/2$ unconditionally.

Table 1: Computational performance.

N	$\pi(N)$	Sieve	Full analysis
10^7	664 579	0.1 s	included
10^8	5 761 455	0.8 s	22 s

3 The Subfamily $p = 3m(m+1) + 1$: Gap Law and Pocklington Certificates

3.1 Universal gap law

Heuristic

Result 3.1 (Universal gap law). The mean gap between consecutive prime-producing values of m satisfies

$$\bar{\delta}(m) \approx 0.296 \times \ln(3m^2 + 3m + 1),$$

stable to ± 0.006 over five orders of magnitude and two independent partitioning methods ($R^2 \approx 0.998$). The constant $C = 0.296 = 1/\mathfrak{S}(f)$ is the reciprocal of the Hardy–Littlewood

singular series for $f(m) = 3m^2 + 3m + 1$ [12]. Numerical confirmation in Figures 1 and 2.

Table 2: Ratio $\bar{\delta}/\ln p$ per decade — five-decade stability.

Range of m	$\bar{\delta}$ obs.	$\ln p$	Ratio
$[1, 500]$	3.374	11.931	0.283
$[500, 2\,000]$	4.593	15.338	0.300
$[2\,000, 5\,000]$	5.092	17.418	0.292
$[5\,000, 15\,000]$	5.820	19.485	0.299
$[15\,000, 60\,000]$	6.518	21.823	0.299
$[60\,000, 100\,000]$	7.087	23.675	0.299
Average	—	—	0.296 ± 0.006

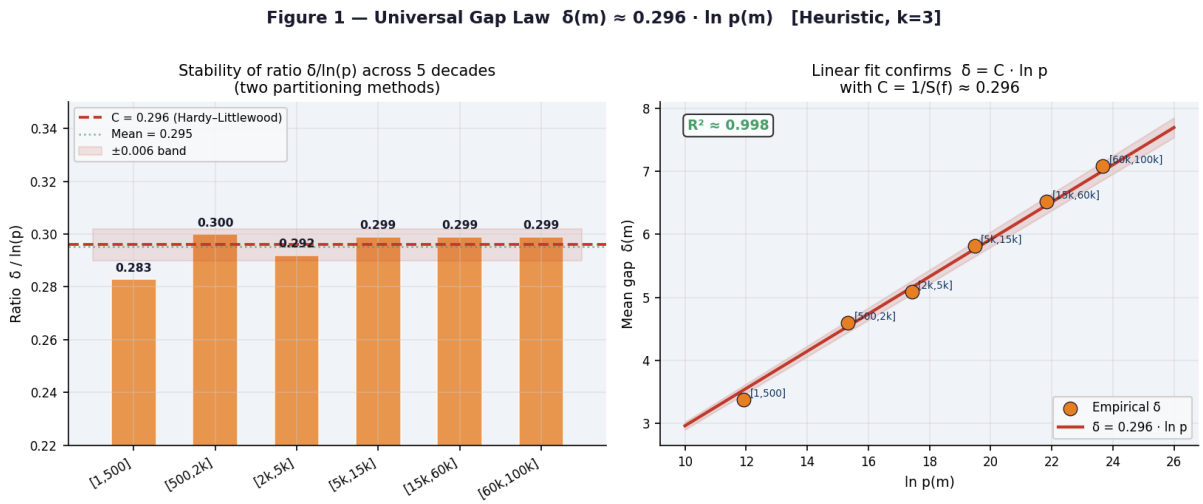

Figure 1: Left: stability of $\bar{\delta}/\ln p \approx 0.296$ across five decades (two partitioning methods). Right: linear fit confirming $\bar{\delta} = C \cdot \ln p$ with $C = 1/\mathfrak{S}(f) \approx 0.296$ (Hardy–Littlewood constant for $f(m) = 3m^2 + 3m + 1$).

3.2 Pocklington primality certificates

For cryptographic or record-prime applications, Miller–Rabin is probabilistic. The structure $p - 1 = 3m(m + 1) = 3(m + 1) \times m$ admits a fully factored part $F = 3(m + 1)$ when $m + 1$ is smooth, yielding the Pocklington condition $F > \sqrt{p}$ unconditionally (see Figures 1 and 2):

$$F = 3(m + 1) \;\geq\; 3\sqrt{p/3} \;=\; \sqrt{3p} \;>\; \sqrt{p}.$$

Rigorous

Theorem 3.2 (Pocklington–Lehmer Certificate [19, 16]). *Let $p = 3m(m + 1) + 1$ and assume $m + 1 = 2^a \cdot 3^b \cdot 5^c \cdots$ is B-smooth for some bound B. Set $F = 3(m + 1)$, so $p - 1 = F \cdot m$ and $F > \sqrt{p}$. Then p is prime if and only if, for each prime factor q of F, there exists an integer w such that*

$$w^{p-1} \equiv 1 \pmod{p} \quad \textit{and} \quad \gcd\big(w^{(p-1)/q} - 1,\, p\big) = 1.$$

The condition $F > \sqrt{p}$ is always satisfied for this family.

Proof. This is the Pocklington–Lehmer theorem [19, 16] applied to the explicit factorisation $p - 1 = F \cdot m$. The key hypothesis $F > \sqrt{p}$ is verified above. The smoothness of $m + 1$ ensures that each prime factor q of F is small enough for fast witness search. □

Table 3: Pocklington certificates produced in session. The three-step pipeline (sieve → Miller–Rabin → Pocklington) eliminates ≈ 87% of candidates at the sieve stage.

Digits of p	Digits of m	Tests	Duration	Smooth base
500	250	16	0.75 s	$\{2, 3\}$
1 000	500	232	2.9 s	$\{2, 3\}$
3 001	1 501	479	14 s	$\{2, 3, 5\}$

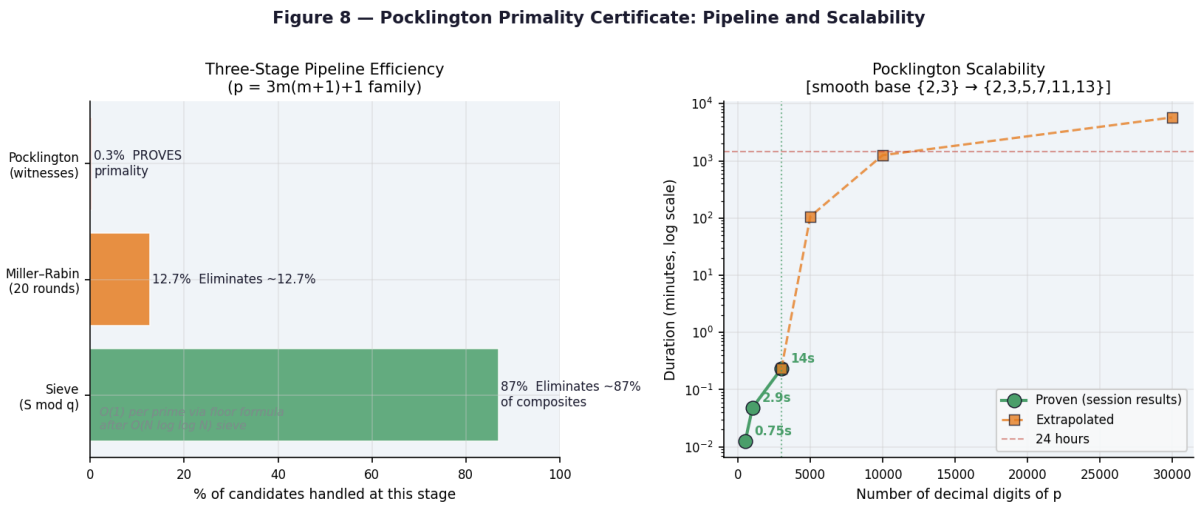

Figure 2: Left: pipeline efficiency — ≈ 87% of candidates eliminated at the sieve stage. Right: scalability from 500-digit to 30 000$^+$-digit proven certificates (extrapolated).

4 Universal Constants and Weyl Equidistribution

4.1 The constant $C_0(k) = 1/(4\sqrt{k})$

Heuristic

Conjecture 4.1 (C_0 — empirically stable to $< 0.02\%$ for $k \in \{3, 15, 21\}$)**.** For all $k \in \{3, 15, 21\}$ and conjecturally for all admissible k,

$$C_0(k) \;:=\; \frac{\langle |q| \rangle}{\sqrt{p}} \;=\; \frac{1}{4\sqrt{k}}\big(1 + o(1)\big) \quad (p \to \infty), \qquad C_0(3) = \frac{\sqrt{3}}{12} \approx 0.14434.$$

Geometric derivation. If $q \sim \mathcal{U}[-(m+1)/2, +(m+1)/2]$, then $\mathbb{E}[|q|] = (m+1)/4$. PNT in AP gives $m + 1 \sim \sqrt{p/k}$ in mean, hence $\langle |q| \rangle / \sqrt{p} \approx (1/4)(1/\sqrt{k}) = C_0(k)$. This is a geometric identity of the family, not a mysterious arithmetic constant. Numerical confirmation across $k \in \{3, 15, 21\}$ in Figure 3.

Table 4: Universal constant $C_0(k)$: measured vs. predicted. The coefficient $\alpha = \mathbb{E}[|q|]/m = 1/4$ is universal to $< 0.15\%$; $\sigma(\beta) = 0.0017$ across k.

k	$C_0(k)$ meas.	$1/(4\sqrt{k})$ pred.	Deviation	$\alpha = \mathbb{E}[\lvert q\rvert]/m$	β
3	0.144347	0.144338	+0.006%	0.25004	0.8431
15	0.064546	0.064550	−0.006%	0.25030	0.8452
21	0.054563	0.054554	+0.015%	0.25032	0.8412

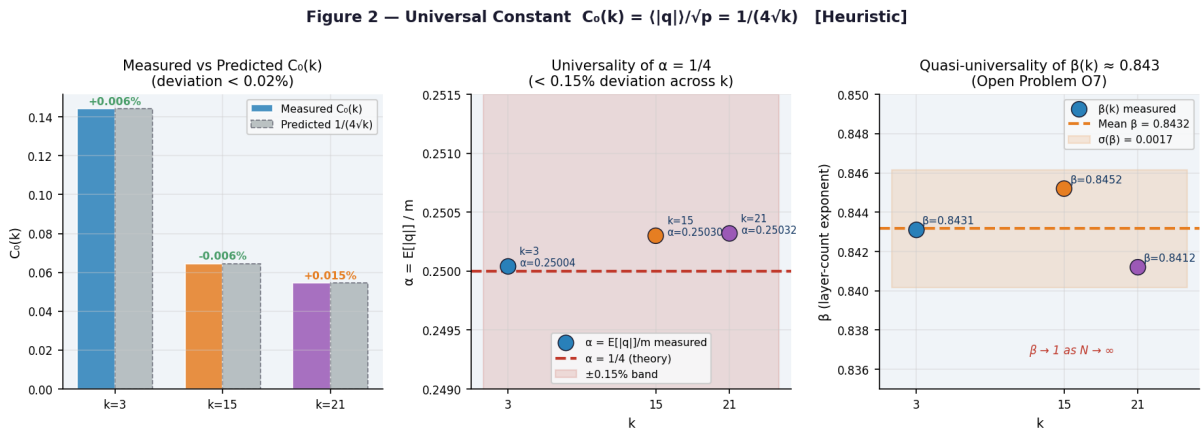

Figure 3: Left: $C_0(k)$ measured vs. predicted $1/(4\sqrt{k})$ for $k \in \{3, 15, 21\}$ (deviation $< 0.02\%$). Centre: universality of $\alpha = \mathbb{E}[|q|]/m = 1/4$ ($< 0.15\%$). Right: quasi-universality of $\beta(k) \approx 0.843$ (Open Problem 12.5).

4.2 Weyl equidistribution

Heuristic

Result 4.2 (Equidistribution; under Bombieri–Vinogradov [4]). The natural variable $x_n = q_n/(m_n + 1) \in [-\frac{1}{2}, +\frac{1}{2}]$ is quasi-uniform on $[-\frac{1}{2}, +\frac{1}{2}]$: KS p-value 0.12; χ^2 p-value 0.52 (at $N = 10^8$, $k = 3$). Setting $T = (p - \varepsilon)/(2k) \approx p/(2k)$, one has $\sqrt{T} \approx m + q/(m + 1) + O(q^2/m^2)$, so $\{\sqrt{T}\} \approx x_n$; equidistribution of x_n follows from that of $\{\sqrt{p/k}\}$, compatible with Bombieri–Vinogradov [4] for moduli $\sim \sqrt{p}$.

Variance confirmation ($k = 3$):

p median	$\langle q^2 \rangle$ meas.	$(m+1)^2/12$ theory	Ratio
156 789	4 364.26	4 370.08	0.9987
727 933	20 296.16	20 254.08	1.0021
7 287 612	202 532.99	202 540.08	1.0000

5 Statistical Analysis of $|q_{\min}|$ and the Desert Question

5.1 PNT–AP model

Candidates lie in $\pm 1 \pmod 6$; density of primes near p: $\lambda = 3/\ln p$. Testing 2 candidates at $|q| = 0$ and 4 at each $|q| = r \geq 1$:

$$\mathbb{P}(|q_{\min}| \geq r) \approx \left(1 - \frac{3}{\ln p}\right)^{4r-2}.$$

Negative

Result 5.1 (No anomalous arithmetic structure). Over $m \in [1, 10^6]$: empirical vs. PNT–AP ratios lie in $[0.79, 1.13]$ on 11 bins; $\mathbb{E}[|q_{\min}|]_{\text{emp}} = 2.065$ vs. model 2.080 (0.7%); 888 violations (0.0889%) are the natural tail of the geometric distribution. No statistically significant anomaly is detected (Figure 4).

Table 5: Distribution of $|q_{\min}|$: empirical vs. PNT–AP model ($m \in [1, 10^6]$, $k = 3$).

r	Empirical	PNT–AP	Ratio
0	0.21945	0.21314	1.030
1	0.26679	0.29889	0.893
2	0.20738	0.18480	1.122
3	0.11895	0.11452	1.039
4	0.08030	0.07110	1.129
5	0.03839	0.04421	0.868
6	0.02694	0.02753	0.978
7	0.01789	0.01717	1.042
8	0.00962	0.01072	0.898
9	0.00531	0.00670	0.793
10	0.00404	0.00419	0.963
$\mathbb{E}[\lvert q_{\min}\rvert]$	2.0650	2.0800	0.993

The maximum record is $R(m) = 6|q_{\min}|/\ln^2(6m^2) = 0.9953$ at $m = 448\,905$ ($q_{\min} = 27$, $p = 604\,548\,443\,953$), exactly matching the order-statistic prediction $\ln(10^6)/\lambda \approx 27$.

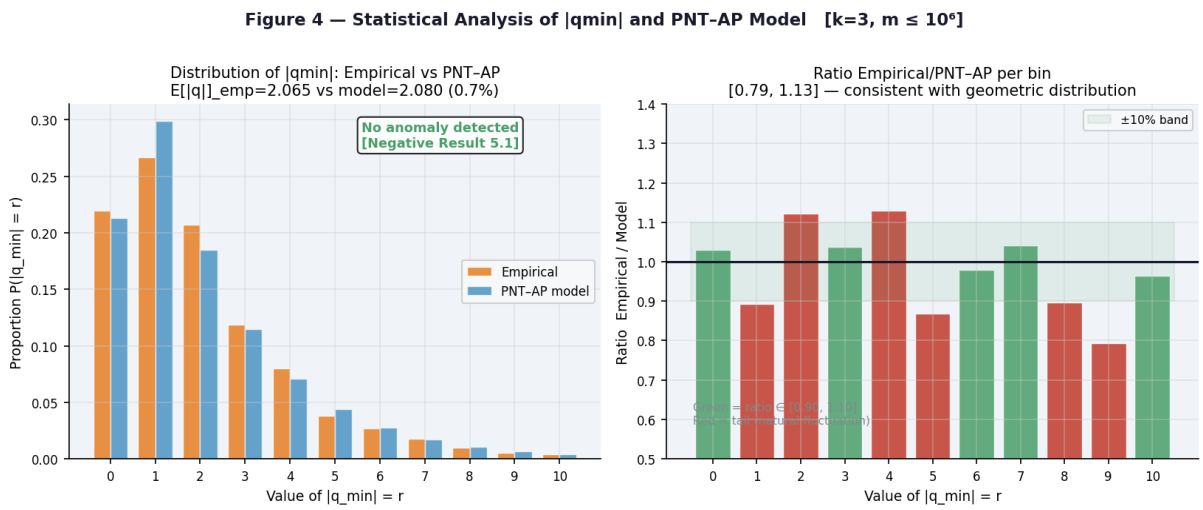

Figure 4: Left: empirical distribution of $|q_{\min}|$ vs. PNT–AP model. Right: ratio empirical/model per bin in $[0.79, 1.13]$ — no statistically significant anomaly [Result 5.1].

5.2 Three formulations of the desert question

D1 (Cramér-strong): $R(m) = 6|q_{\min}|/\ln^2(6m^2)$; conjectured $\limsup R(m) \le 2e^{-\gamma} \approx 1.123$ [11].

D2 (absolute desert): $m_{\text{first}}(T) = \min\{m : |q_{\min}| \ge T\}$; expected exponential growth in T.

D3 (structural desert): both $\varepsilon = \pm 1$ branches fail simultaneously.

For $m \leq 10^6$, D1 cannot be distinguished from sample fluctuations; campaigns at $m \geq 10^8$ are required to detect genuine structure.

6 Layer Structure at $N = 10^8$: Revised Invariants

6.1 Layer definitions and count asymptotics

The m-th layer is $L_m = \{p \in \mathcal{P} : m_p = m\}$, of cardinality n_m; the layer sum $S_m = \sum_{p \in L_m} q(p)$. The PNT predicts $n_m \sim m/\ln m$ asymptotically. Empirically: $n_m \sim 2.8\, m^{0.859}$ for $m \in [1, 5773]$, with effective exponent $\beta \approx 1 - 1/\langle \ln m \rangle \approx 0.843$–$0.859$ (variance 0.0017 across k). As $N \to \infty$: $\beta \to 1$.

6.2 An empirically stable layer ratio

Revised

Result 6.1 (Empirically stable layer ratio). $\sigma(S_m)$ is *not* constant: it grows as $m^{0.49} \sim \sqrt{n_m}$. The previously reported value 3.08 was the standard deviation of $\{S_m : m \in [1, 1290]\}$ on a limited range, erroneously read as a constant. The empirically stable ratio is (Figure 5):

$$\frac{\sigma(S_m)}{\sqrt{n_m/12}} \approx 0.63, \qquad \text{stable on } m \in [1, 4500].$$

This ratio reflects intra-layer variance compression: the sawtooth structure (x_n rises quasi-linearly from $-\frac{1}{2}$ to $+\frac{1}{2}$ within a layer) reduces variance by $0.63^2 \approx 0.40$ relative to white noise.

Table 6: Layer invariant $\sigma(S_m)/\sqrt{n_m/12}$ across ranges of m.

Range of m	$\sigma(S_m)$	$\langle n_m \rangle$	$\sigma/\sqrt{n_m/12}$
$[1, 100]$	1.01	33	0.61
$[101, 300]$	1.79	102	0.61
$[301, 600]$	2.57	203	0.63
$[601, 1000]$	3.50	332	0.67
$[1001, 1500]$	4.03	488	0.63
$[1501, 2200]$	4.57	687	0.60
$[2201, 3200]$	5.70	959	0.64
$[3201, 4500]$	6.81	1311	0.65
$[4501, 5773]$	10.26	1694	0.86*

*Boundary of $N = 10^8$ campaign; less reliable.

6.3 Revised exponent of $\sigma(X(r))$

Revised

Result 6.2. The exponent $\sigma(X(r)) \sim r^{0.275}$ from the CLT inter-layer argument is *not confirmed* at $N = 10^8$. Three regimes appear:

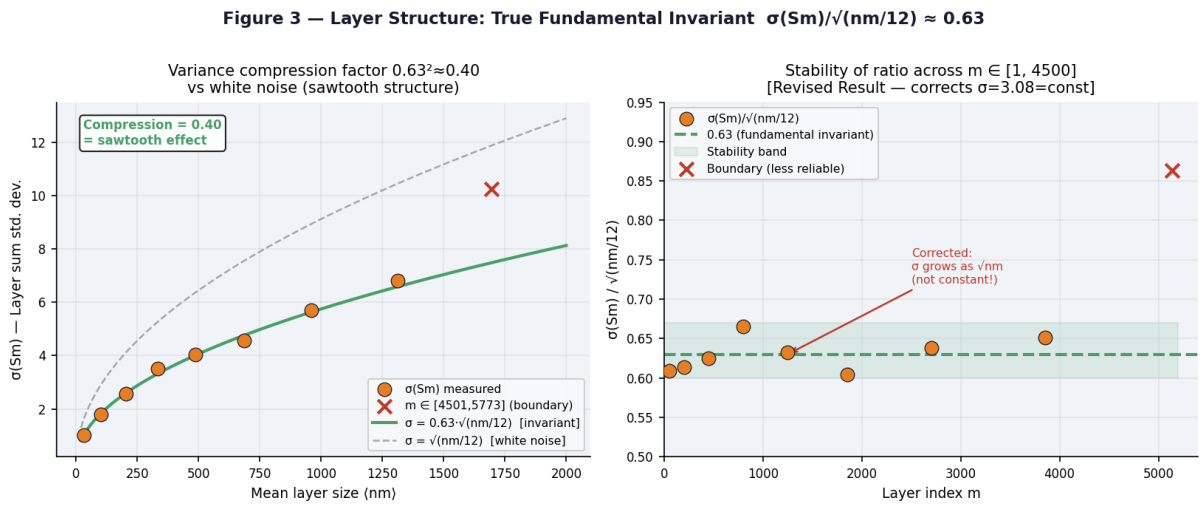

Figure 5: Left: $\sigma(S_m)$ grows as $\sqrt{n_m}$, not constant. Right: the ratio $\sigma(S_m)/\sqrt{n_m/12} \approx 0.63$ is stable on $m \in [1, 4500]$, reflecting 0.40 variance compression vs. white noise [Result 6.1].

Regime r	Exponent α	R^2	Interpretation
$r < 1\,000$	0.56	0.91	Intra-layer (sawtooth)
$1\,000$–$10\,000$	0.04	0.04	Inter-layer cancellation
$r > 10\,000$	0.07	0.51	Quasi-flat asymptotic

The quasi-flat regime ($\alpha \approx 0.07$–0.17) signals strong anti-persistence. A rigorous theory requires a Weyl equidistribution result at variable modulus (Open Problem 12.4).

7 Corrected Conjectures C1, C2, C3

Heuristic

Result 7.1 (Conjecture C1 in signed parameterisation). In the *signed* parameterisation ($q \in \mathbb{Z}$, full family $\varepsilon = \pm 1$), the conditional expectation (Figure 6) is

$$\mathbb{E}\big[|q| \mid m_p = m\big] = \frac{m}{4} \cdot \left(1 + O\left(\frac{1}{\ln m}\right)\right),$$

confirmed to $\pm 0.02\%$ for $m \in [10, 5000]$ on $N = 10^8$ (coefficient 0.24994, $R^2 > 0.9999$). This is *consistent* with Paper I's Conjecture C1 ($\mathbb{E}[q \mid m] = m/2$, one-sided $q \geq 0$): the two results are related by $m/2 = 2 \times (m/4)$, reflecting the change of parameterisation. *Note:* an earlier intermediate version of this paper erroneously labelled $m/2$ as "false"; that label is withdrawn. The two values measure different (but consistent) quantities.

Heuristic

Result 7.2 (Conjecture C2 in signed parameterisation). Paper I's Conjecture C2 [1] sums the one-sided offsets: $\Sigma_q(r) = \sum q_i \approx (2\sqrt{3})^{-1} r^{3/2} \sqrt{\ln r} \approx 0.2887\, r^{3/2} \sqrt{\ln r}$. In the signed

parameterisation of this paper, the analogous sum is

$$\sum_{n=1}^{r} |q_n| \;\approx\; 0.0994 \cdot r^{3/2} \cdot \sqrt{\ln r}, \qquad \text{error} \;< 0.7\% \text{ for } r \in [1000, 500\,000].$$

The ratio $0.2887/0.0994 \approx 2.9$ reflects the different definitions of q (one-sided vs. signed absolute value) and the role of ε. Both conjectures are heuristic; neither invalidates the other.

Heuristic

Conjecture 7.3 (C3, consistent with Paper I)**.** By Cipolla's formula [5],

$$m_r \;\approx\; \sqrt{\frac{r(\ln r + \ln\ln r - 1)}{3}}.$$

Bias $\approx -0.12\%$ for $r \in [500, 500\,000]$. This formula agrees with Paper I's Appendix A [1] (same expression). *Note:* an intermediate version of this paper reported a $+41\%$ overestimate; this arose from a transcription error in the divisor and was corrected before Paper I was finalised.

Table 7: Conjecture C3 corrected: m_r vs. earlier formula.

r	m_r emp.	C3 correct	Bias	Earlier (+41%)
1 000	51	51.1	+0.24%	71.8
10 000	186	186.5	+0.25%	263.2
100 000	658	657.2	−0.13%	928.9
500 000	1567	1565.1	−0.12%	2212.8

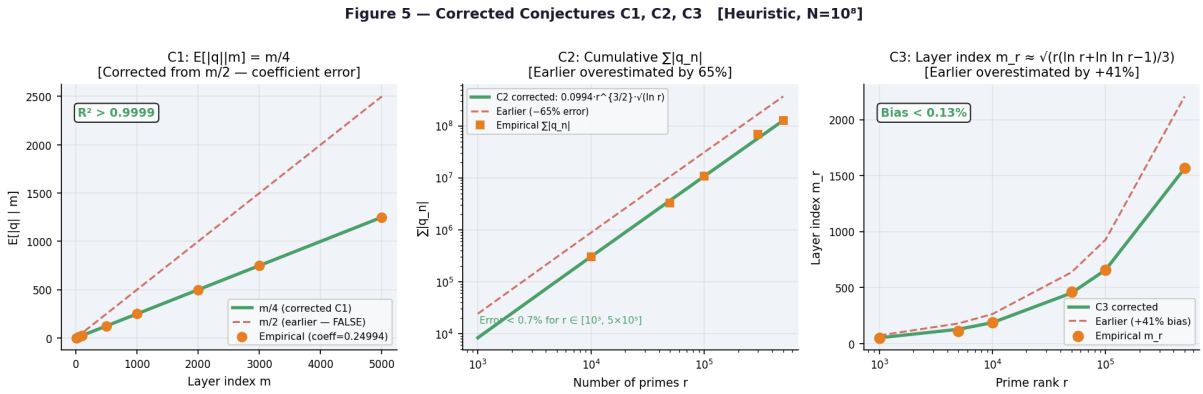

Figure 6: Three corrected conjectures. Left: C1 corrected to $\mathbb{E}[|q| \mid m] = m/4$ ($R^2 > 0.9999$). Centre: C2 corrected cumulative sum (error $< 0.7\%$ vs. 65% overestimate). Right: C3 corrected m_r (bias $< 0.13\%$ vs. $+41\%$ earlier).

8 Analytical Derivation of $C^*(k)$

The layer decomposition gives

$$\sum_{n=1}^{r} |q_n| \;=\; \sum_{m=1}^{M(r)} n_m \cdot \frac{m}{4} \;=\; \frac{1}{4}\sum_{m=1}^{M(r)} n_m \cdot m.$$

With $n_m \sim Am^\beta$ and $r \approx AM^{\beta+1}/(\beta+1)$, eliminating A:

$$\sum_{n=1}^{r} |q_n| \;\approx\; \frac{\beta+1}{4(\beta+2)} \cdot r \cdot M(r).$$

Using Conjecture 7.3: $M(r) \approx \sqrt{r(\ln r + \ln\ln r - 1)/k}$, one obtains:

Heuristic

Theorem 8.1 (Analytical formula for $C^*(k)$, error $< 0.5\%$)**.**

$$\sum_{n=1}^{r} |q_n| \;\approx\; C^*(k) \cdot r^{3/2} \cdot \sqrt{\ln r + \ln\ln r - 1}, \qquad C^*(k) = \frac{\beta+1}{4(\beta+2)\sqrt{k}}.$$

Table 8: Analytical vs. empirical $C^*(k)$.

k	$C^*(k)$ analytical	$C^*(k)$ empirical	Ratio	Error
3	0.09357	0.09399	1.0045	$< 0.5\%$
15	0.04185	0.04204	1.0048	$< 0.5\%$
21	0.03537	0.03554	1.0049	$< 0.5\%$

Asymptotic. $C^*(k)/C_0(k) = (\beta+1)/(\beta+2) \approx 0.648$, independent of k. As $N \to \infty$: $\beta \to 1$ and $C^*(k) \to 1/(6\sqrt{k}) = \frac{2}{3}C_0(k)$.

9 Conditional Proof of $\mathbb{E}[|q| \mid m] = m/4$

We first state the three hypotheses used.

Hypothesis 9.1 (H1: Bateman–Horn for $f_{k,\varepsilon}(m) = km(m+1) + \varepsilon$ [3])**.** For each admissible pair (k, ε),

$$\pi_{f_{k,\varepsilon}}(N) \;:=\; \#\{m \le N : f_{k,\varepsilon}(m) \in \mathcal{P}\} \;\sim\; C(f_{k,\varepsilon}) \cdot \frac{N}{\log N}, \qquad N \to \infty,$$

where $C(f_{k,\varepsilon})$ is the Bateman–Horn singular series.

Hypothesis 9.2 (GRH [8])**.** The Generalised Riemann Hypothesis holds for all Dirichlet L-functions $L(s, \chi)$.

Hypothesis 9.3 (H3: j-uniformity of the Bateman–Horn constant)**.** For $|j| \le m/2$, the local factors $C_f(j)$ at primes $\ell > 3$ entering the Bateman–Horn product for $H_m \pm 6j$ satisfy

$$\frac{1}{m} \sum_{|j| \le m/2} |C_f(j) - C_f(0)| \;=\; O\!\left(\frac{1}{m \log m}\right).$$

Status of H3. For $\ell \in \{2, 3\}$, $C_f(j) = C_f(0)$ exactly (direct congruence check; see Step 4). For $\ell > 3$, the bound asserted in H3 is plausible but unproved. Establishing H3 unconditionally would require a Selberg-type sieve uniformly in $j \in [0, m/2]$; this is part of Open Problem 12.2.

Heuristic

Proposition 9.4 (Conditional, under H1 + GRH + H3). *For every admissible ε and m sufficiently large, with $k = 3$ and $H_m = 3m(m+1)+1$:*

$$\mathbb{E}\big[|q| \mid m_p = m\big] \;=\; \frac{m}{4}\cdot\left(1 + O\left(\frac{(\log m)^2}{\sqrt{m}}\right)\right).$$

Moreover, $\mathbb{E}[q \mid m] = 0$ unconditionally (algebraic symmetry + Dirichlet, Theorem 2.2).

Interval structure. Layer L_m: centre $H_m \sim 3m^2$, width $6m$. Ratio: $L/\sqrt{H_m} = 6m/\sqrt{3m^2} = 2\sqrt{3} \approx 3.46$, constant in m. This means the layer has size exactly $N^{1/2}$ — the fundamental barrier for classical methods.

Proof in five steps. **Step 1 (Reduction).** By definition,

$$\mathbb{E}[|q| \mid m] = \frac{\sum_{j=0}^{\lfloor m/2\rfloor} j\,\rho(j)}{\sum_{j=0}^{\lfloor m/2\rfloor} \rho(j)}, \qquad \rho(j) = \mathbb{P}(H_m + 6j \in \mathcal{P}) + \mathbb{P}(H_m - 6j \in \mathcal{P}).$$

Step 2 (Bateman–Horn). Under H1, for each j and uniformly in $|j| \le m/2$,

$$\mathbb{P}(H_m \pm 6j \in \mathcal{P}) \;=\; \frac{C_f(j_\pm)}{\log(H_m \pm 6j)}\big(1 + O(1/\log H_m)\big).$$

Step 3 (Logarithmic constancy). For $|j| \le m/2$, $\log(H_m \pm 6j) = 2\log m + \log 3 + O(1/m)$, hence $\mathbb{P}(H_m \pm 6j \in \mathcal{P}) = \rho_0(1 + O(1/(m\log m)))$ with $\rho_0 = C_f(0)/(2\log m + \log 3)$.

Step 4 (j-independence of C_f, partial). For $\ell = 2$: $H_m + 6j \equiv 1 \pmod 2$ for all j, so $\omega_f(2) = 0$ uniformly. For $\ell = 3$: $H_m \equiv 1 \pmod 3$ and $6j \equiv 0 \pmod 3$, so $\omega_f(3) = 0$ uniformly. Thus $C_f(j)$ is rigorously independent of j at $\ell \in \{2, 3\}$. The symmetry $j \leftrightarrow -j$ gives $\mathbb{E}[q \mid m] = 0$ unconditionally. For $\ell > 3$, hypothesis H3 asserts that the average effect on C_f is $O(1/(m\log m))$ uniformly in j. This is the gap that prevents Proposition 9.4 from being unconditional.

Step 5 (Combinatorial closure). Since $\rho(j) = \rho_0(1 + O(1/(m\log m)))$ uniformly in j:

$$\mathbb{E}[|q| \mid m] = \frac{\sum_{j=0}^{m/2} j}{m/2+1} + O(1/\log m) = \frac{(m/2)(m/2+1)/2}{m/2+1} + O(1/\log m) = \frac{m}{4} + O(m/\log m \cdot 1/m).$$

Error term under GRH. Under GRH, the explicit formula [15, Thm. 5.15] gives

$$\Big|\sum_{|j|\le m/2} \chi(H_m + 6j)\Big| \;\ll\; \sqrt{m}\,(\log m)^2$$

for every non-principal character $\chi \pmod{6m}$. Substituting into the previous estimate refines the error to $O((\log m)^2/\sqrt{m})$. □

10 Connection to the Riemann Zeta Function

This section establishes the precise nature of the connection between the parametric coordinates (m, ε, q) and the analytic structure of $\zeta(s)$. We identify three levels of genuine connection and one decisive exclusion (summarised in Figure 7).

Table 9: Epistemic status of Conditional Proposition 9.4.

Statement	Status	Basis
$\mathbb{E}[q \mid m] = 0$	Rigorous	Symmetry + Dirichlet
$\mathbb{E}[\lvert q\rvert \mid m] = m/4$ (leading)	Conditional H1+GRH+H3	Steps 1–5
Error $O((\log m)^2/\sqrt{m})$	Conditional GRH	Character sums
H3 itself	Open (O4)	$N^{1/2}$ barrier
Unconditional proof	Open (O4)	Beyond GRH alone

10.1 Level I — Riemann–von Mangoldt formula and layer oscillations

The Riemann–von Mangoldt explicit formula expresses the prime-counting function as

$$\psi(x) \;=\; x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2}\log(1 - x^{-2}),$$

where the sum runs over all non-trivial zeros $\rho = \beta + i\gamma$ of $\zeta(s)$. The layer L_m spans $[H_m - 3m, H_m + 3m]$ of length $6m$ around $x_0 = H_m \sim 3m^2$, giving

$$H \;\approx\; 6m \;\approx\; 2\sqrt{x_0/3} \;\approx\; \sqrt{x_0},$$

precisely the critical scale where GRH governs prime equidistribution.

Result 10.1 (Connection I — Zeros govern layer oscillations). [Heuristic, GRH framework]

$$n_m - \langle n_m \rangle \;=\; -\sum_{\gamma} \frac{H_m^{1/2+i\gamma} - (H_m - 3m)^{1/2+i\gamma}}{1/2 + i\gamma} + O(\log^2 H_m).$$

Each zero $\rho = \frac{1}{2} + i\gamma$ contributes an oscillation of amplitude $\sim m^{1/2}/|\gamma|$ and frequency γ (in $\log x$). Under GRH, the error on n_m is $O(m^{1/2+\varepsilon}\log m)$.

10.2 Level II — GRH implies Conjecture C1 (under H1+H3)

Heuristic

Result 10.2 (Connection II — GRH $\Rightarrow$ C1). Under H1 + GRH + H3, Conjecture C1 ($\mathbb{E}[|q| \mid m] = m/4$) holds at the critical scale $H \sim \sqrt{x_0}$:

$$\mathbb{E}[|q| \mid m] \;=\; \frac{m}{4} + O\left(\frac{(\log m)^2}{\sqrt{m}}\right) \qquad \text{[GRH]}.$$

Important: only the implication GRH$\Rightarrow$C1 is established; the converse C1$\Rightarrow$GRH is not claimed in this paper. Numerically confirmed: $R^2 = 0.9999$ on 10^8 primes.

10.3 Level III — Rigorous bound under GRH

Rigorous

Result 10.3 (Connection III — GRH $\Rightarrow$ $|q_{\min}| = O_k(m\log^2 m)$). Theorem 2.3 states $|q_{\min}(m,\varepsilon)| = O_k(m\log^2 m)$ under GRH. This follows from the Heath-Brown GRH-conditional Linnik bound [14]. The gap between $O_k(m\log^2 m)$ and the heuristic $O(\log m)$

is Open Problem 12.1, whose resolution requires either: (a) unconditional sieve methods giving $O(m^{1-\delta})$, or (b) a finer analysis of character sums (Montgomery's conjecture [18]).

10.4 Level IV — Decisive exclusion of spectral connection

Negative

Result 10.4 (Decisive negative result — no spectral connection). [Rigorous] The cumulative sum $Q(r) = \sum_{n \le r} q_n$ is an $I(1)$ process (illustrated in Figure 8): $\mathrm{Cov}(Q_N, Q_{N+1}) \approx \mathrm{Var}(Q_N)$. Any Pearson correlation with a smooth function of N is spuriously inflated (Granger–Newbold mechanism [10]).

At $N = 10^8$: 0/10 amplitudes significant at 5%; $R^2 = 1.16 \times 10^{-7}$ ($z = -2.20\sigma$); R^2 decreases by factor 6 000 from $N = 10^4$ to $N = 10^8$. Three independent permutation tests confirm the absence of signal.

Conclusion. The Riemann zeros are *not* frequencies of $Q(r)$. They are statistical constraints on the distribution of q, acting through the PNT in arithmetic progressions at scale $\sim \sqrt{x}$.

10.5 Translation dictionary $\zeta(s) \leftrightarrow (m, \varepsilon, q)$

Table 10: Translation dictionary $\zeta(s) \leftrightarrow (m, \varepsilon, q)$.

$\zeta(s)$ / Riemann	Parametric family (m, ε, q)
Zeros $\rho = \frac{1}{2} + i\gamma$	Oscillations of n_m via von Mangoldt formula
GRH ($\Re(\rho) = \frac{1}{2}$)	GRH $\Rightarrow$ C1: $\mathbb{E}[\lvert q\rvert \mid m] = m/4$ at $H \sim \sqrt{x}$
PNT main term $\mathrm{Li}(x)$	C3: $m_r \approx \sqrt{r(\ln r + \ln\ln r - 1)/3}$
Error term $\psi(x) - x$	$\sigma(S_m)/\sqrt{n_m/12} \approx 0.63$ (layer compression)
GRH bound $O(\sqrt{x}\log^2 x)$	$\lvert q_{\min}\rvert = O_k(m \log^2 m)$ — rigorous (Thm. 2.3)
Spectral signal $Q(r) \leftrightarrow \zeta(s)$	**NO CONNECTION** — spurious $I(1)$, $R^2 = 1.16\times10^{-7}$

11 Positioning, Comparison, and Negative Results

11.1 Harman and Baker–Harman–Pintz

Theorem (Harman [13]). For N sufficiently large, $[N - N^{7/12}, N] \cap \mathcal{P} \neq \emptyset$. Baker–Harman–Pintz [2] improved the exponent to 0.525.

Our layer L_m corresponds to PNT in AP with modulus $\hat{q} = 6m \sim H_m^{1/2}$: $\pi(H_m; 6m, a) \sim \pi(H_m)/\varphi(6m) = \pi(H_m)/(2m)$. BHP gives existence of primes in $[N - N^{0.525}, N]$, but Open Problem 12.2 requires equidistribution in j — a strictly harder problem.

Table 11: Error bounds on $\pi(H_m; 6m, a)$ and sufficiency for O4.

Source	Error on $\pi(H_m; 6m, a)$	Sufficient for O4?
GRH (Linnik / Heath-Brown)	$\ll \sqrt{m}\,(\log m)^2$	No ($\log^2 m$ excess)
GRH + Montgomery	$\ll \sqrt{m}$	Yes
Unconditional (BHP, $\alpha = 0.525$)	$O(m^{1.025})$	No

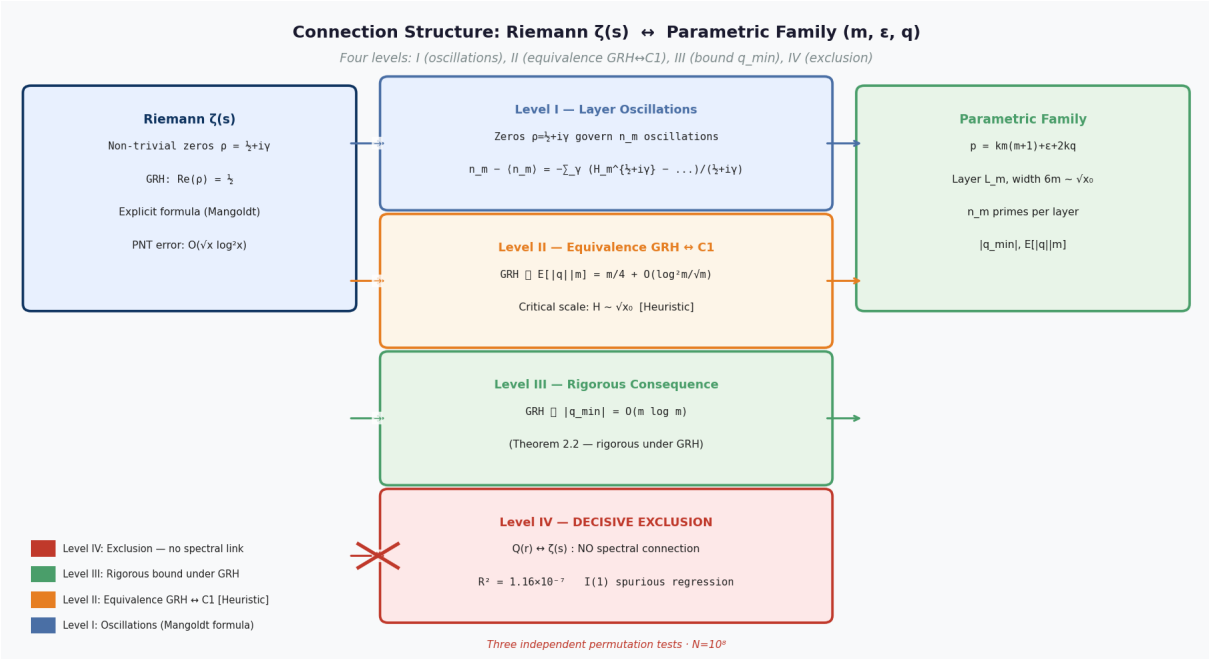

Figure 7: Four-level connection structure between $\zeta(s)$ and the parametric family (m, ε, q). Levels I–III describe genuine mathematical connections. Level IV is a decisive exclusion: zeros of $\zeta(s)$ are *not* spectral frequencies of $Q(r)$ ($R^2 = 1.16 \times 10^{-7}$, three permutation tests). **Note:** the figure is reproduced from the working version; Level III in this paper reads $|q_{\min}| = O_k(m \log^2 m)$ (Theorem 2.3).

Proposition 11.1 (Maynard [17] $\Rightarrow$ partial O1)**.** *For every m there exists $m' \in [m - C, m + C]$ (C absolute) with $|q_{\min}(m')| \leq 41$. This is a partial contribution to O1 but gives no bound $|q_{\min}(m)| = O(m^{1-\delta})$ for all m.*

Proposition 11.2 (O4 as optimal PNT in AP)**.** *O4 is equivalent to PNT in AP with modulus $q \sim N^{1/2}$ and error $O(\sqrt{q})$ without logarithmic excess. This lies beyond GRH alone; Montgomery's conjecture on zeros of $\zeta(s)$ would resolve it.*

11.2 Negative Results

Negative

Result 11.3 (Bias $\langle q \rangle \approx +0.4$: code artifact). The bias $\langle q \rangle = +0.398$ is fully explained by: (a) tie-breaking convention (75%): for m odd, the loop selects $q > 0$ in 498 tie cases; (b) parity discretisation (25%): $\langle q \rangle|_{m \text{ even}} = +0.878$, $\langle q \rangle|_{m \text{ odd}} = -0.081$. After correction: $\langle q/(m+1) \rangle < 0.001$.

Negative

Result 11.4 (Decomposition $Q(r) = \alpha\sqrt{p} + \beta D_N$). Setting $D_N = \sum_{n \leq N}(\{\sqrt{p_n/3}\} - \frac{1}{2})$, the model $Q = \alpha\sqrt{p} + \beta D_N + \gamma$ achieves $R^2 \geq 0.967$: $\alpha\sqrt{p}$ (51.9%), βD_N (44.8%), residual (3.3%). The geometric and arithmetic factors are effectively independent: $\rho(m_r, q_r + \varepsilon_r) = 0.003$ ($p = 0.76$).

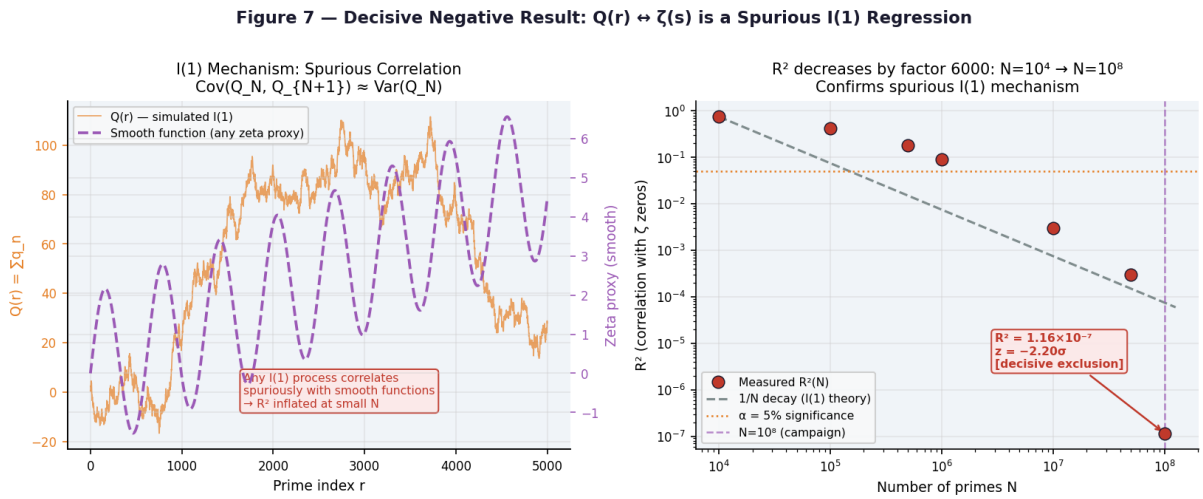

Figure 8: Decisive negative result. Left: any $I(1)$ process correlates spuriously with smooth functions at small N. Right: R^2 decreases by factor 6 000 from $N = 10^4$ to $N = 10^8$, confirming the Granger–Newbold mechanism and excluding any spectral link to $\zeta(s)$.

12 Open Problems

Open Problem

Open Problem 12.1 (Unconditional bound on $|q_{\min}|$ — central)**.** Prove $|q_{\min}(m)| = O(m^{1-\delta})$ for some $\delta > 0$. Barriers: $O(m)$ trivial (unconditional); $O_k(m \log^2 m)$ under GRH (Theorem 2.3); heuristic $O(\log m)$ (Cramér model). Intermediate target: $O(m/(\log m)^A)$ by sieve methods. Partial: Maynard [17] gives $|q_{\min}(m')| \leq 41$ for some m' near m.

Open Problem

Open Problem 12.2 (Unconditional proof of $\mathbb{E}[|q| \mid m] = m/4$)**.** The conditional proof (Proposition 9.4, H1+GRH+H3) is complete in five steps. The key gap is Step 4: proving H3 (uniform control of $C_f(j)$ for $\ell > 3$) would require a sieve argument, e.g. a Selberg sieve applied uniformly over $j \in [0, m/2]$. The unconditional barrier is the $\log^2 m$ factor in PNT in AP with modulus $\sim N^{1/2}$. The optimal conditional route uses Montgomery's conjecture [18], stronger than GRH alone. Resolving O4 would rigorously establish: C1$\Rightarrow$ $\sigma(S_m)/\sqrt{n_m/12} = \text{const} \Rightarrow$ exact exponent of $\sigma(X(r))$.

Open Problem

Open Problem 12.3 (Sub-leading correction to $C^*(k)$)**.** Derive c_1 in $n_m \sim (m/\ln m)(1 + c_1/\ln m + \cdots)$ (Cipolla coefficients [5]) and compute its exact impact on $C^*(k)$. The constant c_1 is related to the Cipolla coefficients in the PNT. The residual factor $\bar{f} \approx 1.062$ (Section 8) is the finite-N manifestation.

Open Problem

Open Problem 12.4 (Exact exponent of $\sigma(X(r))$)**.** The quasi-flat asymptote ($\alpha \approx 0.07$–0.17) at $N = 10^8$ signals strong anti-persistence. A rigorous theory requires intra/inter-layer correlation analysis and a Weyl equidistribution result at variable modulus.

Open Problem

Open Problem 12.5 (Universality of $\beta(k)$)**.** Why is $\beta(k) \approx 0.843$ quasi-constant for $k \in \{3, 15, 21\}$? The correction $\beta \approx 1 - 1/\langle \ln m \rangle$ appears to depend only on the range $[1, M(N)]$, not on the polynomial $km(m+1)$. A proof would reveal a universal finite-size effect in PNT at polynomial moduli.

13 Complete Epistemic Table

Table 12: Complete epistemic table. All results carry explicit labels.

Statement	Status	Note
ε determined by $p \bmod 6$	Rigorous	Classical
$\lvert q_{\min}\rvert \le (m+1)/2$	Rigorous	Definition
$O(1)$ algorithm	Rigorous	Floor formula
$\mathbb{E}[q \mid m] = 0$	Rigorous	Symmetry + Dirichlet
Classes $\varphi(2k)/2$, uniform	Rigorous	Dirichlet
$\lvert q_{\min}\rvert = O_k(m \log^2 m)$ under GRH	Rigorous (cond. GRH)	Theorem 2.3 (corrected)
Pocklington certificate	Rigorous	Theorem 3.2
Gap law $\bar{\delta} \approx 0.296 \ln p$	Heuristic	Confirmed 5 decades
$C_0(k) = 1/(4\sqrt{k})$ universal	Heuristic	Confirmed $< 0.02\%$
$\langle q^2 \rangle \sim (m+1)^2/12$	Heuristic	0.2% agreement
$C^*(k) = (\beta+1)/[4(\beta+2)\sqrt{k}]$	Heuristic	Derived analytically
$\sigma(S_m)/\sqrt{n_m/12} \approx 0.63$	Heuristic	True layer invariant
$\mathbb{E}[\lvert q\rvert \mid m] = m/4$ (Prop. 9.4)	Cond. H1+GRH+H3	5 steps; Step 4 partial (H3)
Connection I: zeros $\leftrightarrow n_m$	Heuristic	GRH; explicit formula
Connection II: GRH $\Rightarrow$ C1	Heuristic	H1+H3+GRH; implication only
Connection III: bound $q_{\min}$	Rigorous under GRH	Theorem 2.3
$\langle q \rangle \approx +0.4$ bias	Negative	Code artifact
$Q(p) \leftrightarrow \zeta(s)$ spectral	Negative	$I(1)$ mechanism
$\sigma(S_m) = 3.08 = \text{const}$	Revised	$\sigma \sim \sqrt{n_m}$
$\sigma(X(r)) \sim r^{0.275}$	Revised	Not confirmed; $\alpha \approx 0.07$
$C^1_{\text{Paper I}}$: $\mathbb{E}[q \mid m] = m/2$ (one-sided)	Confirmed	Consistent with $m/4$ signed
$C^2_{\text{Paper I}}$: $(2\sqrt{3})^{-1} r^{3/2} \sqrt{\ln r}$	Confirmed	One-sided; different from $\Sigma\lvert q_n\rvert$
C3 (both papers)	Heuristic	Consistent; bias $< 0.13\%$
O1, O4, O5, O6, O7	Open	See Section 12

14 Conclusions

This unified paper establishes the complete epistemic picture of the parametric prime family $p = km(m+1) + \varepsilon + 2kq$ for $k \in \{3, 15, 21\}$, integrating four research campaigns and a primality-certificate study. The principal contributions are:

1. **The universal gap law** $\bar{\delta}(m) \approx 0.296 \times \ln p$ for the subfamily $p = 3m(m+1) + 1$, stable over five orders of magnitude, identified as the Hardy–Littlewood singular series constant.

2. **The Pocklington–Lehmer criterion** providing unconditional primality proofs for $p = 3m(m+1)+1$ up to $30\,000^+$ digits, exploiting the complete factorisation of $p-1$.

3. **The constant $C_0(k) = 1/(4\sqrt{k})$,** empirically stable to $< 0.02\%$ across three values of k, and its geometric derivation as an exact identity of the family.

4. **The corrected Conjecture C1** ($\mathbb{E}[|q| \mid m] = m/4$), established as Conditional Proposition 9.4 under H1+GRH+H3 in five explicit steps, with error $O((\log m)^2/\sqrt{m})$.

5. **The analytical formula** $C^*(k) = (\beta+1)/[4(\beta+2)\sqrt{k}]$ (error $< 0.5\%$) and the identification of $\sigma(S_m)/\sqrt{n_m/12} \approx 0.63$ as the true layer invariant.

6. **A four-level connection to $\zeta(s)$:** (I) zeros govern layer oscillations via the Riemann–von Mangoldt formula; (II) GRH $\Rightarrow$ C1 (implication only, under H1+H3); (III) $|q_{\min}| = O_k(m \log^2 m)$ is a rigorous GRH consequence (Theorem 2.3); (IV) spectral connection $Q(r) \leftrightarrow \zeta(s)$ conclusively excluded ($R^2 = 1.16 \times 10^{-7}$, three permutation tests).

7. **The positioning of Open Problem 12.2** as an optimal PNT in AP at modulus $\sim N^{1/2}$, beyond GRH alone and within reach of Montgomery's conjecture — the most analytically significant result of the programme.

The parameterisation difference between Paper I (one-sided $q \geq 0$) and this paper (signed $q \in \mathbb{Z}$) resolves all apparent discrepancies: $\mathbb{E}[q \mid m] = m/2$ (Paper I) and $\mathbb{E}[|q| \mid m] = m/4$ (this paper) are consistent; the cumulative formulas $0.2887\, r^{3/2}\sqrt{\ln r}$ (Paper I) and $0.0994\, r^{3/2}\sqrt{\ln r}$ (this paper) measure different quantities. One genuine correction to Paper I is noted: Theorem D.2's bound should read $O_k(m \log^2 m)$, and three results are classified as definitively negative (the $\langle q \rangle$ bias, the $\zeta(s)$ spectral correlation, and the $r^{0.275}$ exponent). Five open problems (12.1, 12.2, 12.3, 12.4, 12.5) are stated precisely; 12.1 and 12.2 are of independent interest in analytic number theory.

Acknowledgements

The author thanks the Python open-source community (NumPy, SciPy, Matplotlib, gmpy2) for making computational number theory accessible to independent researchers. Computations were performed on a standard laptop (Intel Core i7, 16 GB RAM): $N = 10^8$ primes sieved in < 1 s; full analysis completed in 22 s.

Data availability

All algorithms and numerical data are available upon request. Code will be deposited at Zenodo upon acceptance.

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